(ii) Find $\left(\frac{N}{2}\right)^{th}$ term.

(iii) The class that contains the cumulative frequency $\frac{N}{2}$ is called the median class.

(iv) Find the median by using the formula:

$$\text{Median} = l + \frac{\left(\frac{N}{2} - m\right)}{f} \times c$$

Where l = *Lower limit of the median class,* f = *Frequency of the median class*

c = *Width of the median class,* N = *The total frequency* $\left(\sum f\right)$

m = *cumulative frequency of the class preceeding the median class*

Example 8.12 The following table gives the weekly expenditure of 200 families. Find the median of the weekly expenditure.

Weekly expenditure (₹)	0-1000	1000-2000	2000-3000	3000-4000	4000-5000
Number of families	28	46	54	42	30

Solution

Weekly Expenditure	Number of families (f)	Cumulative frequency (cf)
0-1000	28	28
1000-2000	46	74
2000-3000	54	128
3000-4000	42	170
4000-5000	30	200
	N=200	

Median class $= \left(\frac{N}{2}\right)^{\text{th}}$ value $= \left(\frac{200}{2}\right)^{\text{th}}$ value

$= 100^{\text{th}}$ value

Median class $= 2000 - 3000$

$\frac{N}{2} = 100 \quad l = 2000$

$m = 74, \quad c = 1000, \quad f = 54$

$$\text{Median} = l + \frac{\left(\frac{N}{2} - m\right)}{f} \times c$$

Progress Check

1. The median of the first four whole numbers _____.
2. If 4 is also included to the collection of first four whole numbers then median value is _____.
3. The difference between two median is _____.

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